



CS 03500 - Foundations of Cybersecurity

Cyberwarfare: Information Operations in a Connected World Projects

Project Part 2:

Identifying Cyber Threats and Applying the Cyber Kill Chain

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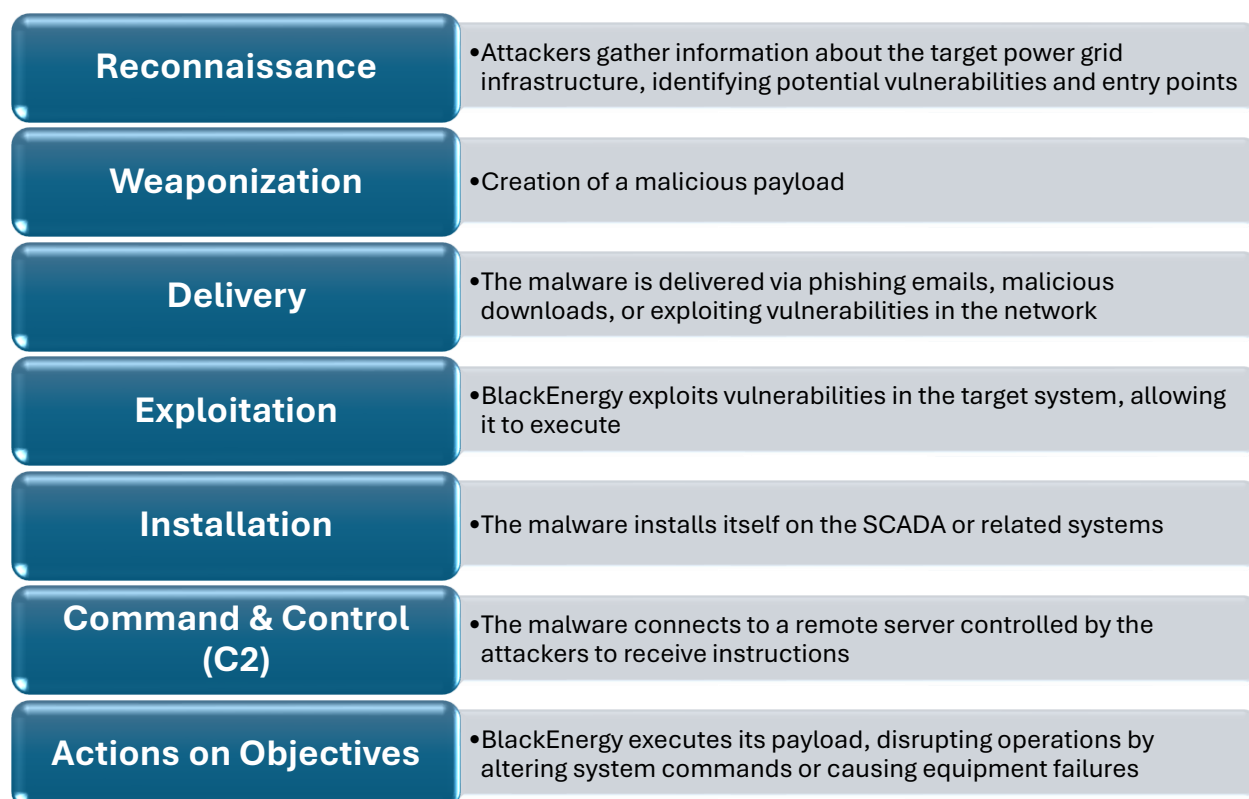
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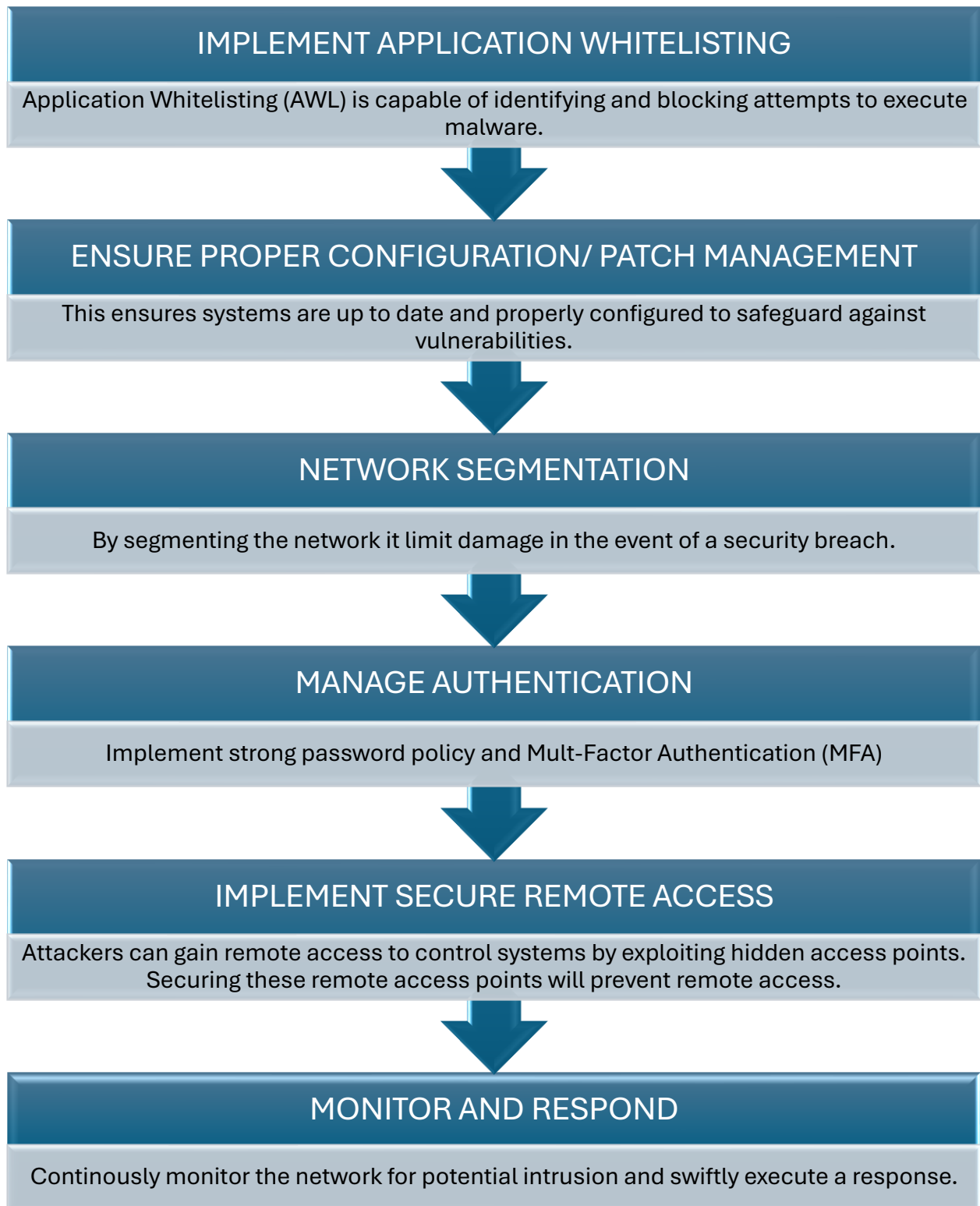
A specific type of malware that could affect the Western Interconnection power grid is BlackEnergy which is a type of Trojan malware employed for Distributed Denial of Service (DDoS) attacks, cyber espionage, and data destruction targeting SCADA (Supervisory Control and Data Acquisition) systems. BlackEnergy can be installed through phishing emails or exploiting vulnerabilities such as unpatched software or weak security configurations in the SCADA, thus allowing attackers to disrupt the network by manipulating control commands, leading to equipment failures, altering operational settings, or launching DDoS attacks to overwhelm network resources, thereby crippling critical operations and gain control over power grid operations.

The Cyber Kill Chain model consists of reconnaissance, weaponize, deliver, exploit, install, command and control and act on objectives (Chapple and Seidl 114–22).

A detailed description of how it applies to Blackenergy malware is outlined below.



To eradicate BlackEnergy from the network, the following actions should be taken (ICS-CERT):



To apply the Adversary Model to the BlackEnergy cyberattack group the following elements should be considered capabilities, resources, intent, motivation, access and risk aversion (Mullins):



Works Cited

“BlackEnergy APT Attacks | What is BlackEnergy? | Threat Definition.” /, 20 Nov. 2017,

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Chapple, Mike, and David Seidl. *Cyberwarfare: Information Operations in a Connected*

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Mullins, Robert. “The Cold War: Then and now.” *Network World*, 21 Nov. 2010,

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